Long-Fiber Single-Mode Raman Suppression

Fiber Raman Suppression Project

Evidence snapshot: 2026-04-28

Abstract

This note summarizes the single-mode long-fiber Raman-suppression lane. The main result is that the project can run checkpointed 100–200 m SMF-style optimizations and produce image-backed suppression near −55 dB from almost unshaped Raman transfer. The important caveat is that these are completed long-fiber milestones, not converged global optima. The current value of the lane is practical and scientific: it shows how short-fiber phase structure can seed much longer propagation, how the grid/checkpoint workflow must change, and where the remaining convergence and lab-readiness gaps are.

Claim Status

Status: completed milestone with convergence caveats
The 100 m and 200 m single-mode result artifacts have complete standard images and strong scalar suppression. They should be presented as supported exploratory long-fiber simulation results, not as final optimized lab waveforms.

1 Question and Thesis

The short-fiber baseline asks whether input spectral phase can suppress Raman transfer over a few meters. The long-fiber question is harder:

Can the same phase-only control idea still work over 100–200 m?

The answer is yes, with caveats. The result is not a clean theorem that one phase profile works for every long fiber. It is a computational finding: with explicit long-fiber grids, warm starts, checkpointed L-BFGS, and standard image inspection, the project can reach deep Raman suppression in 100–200 m single-mode simulations.

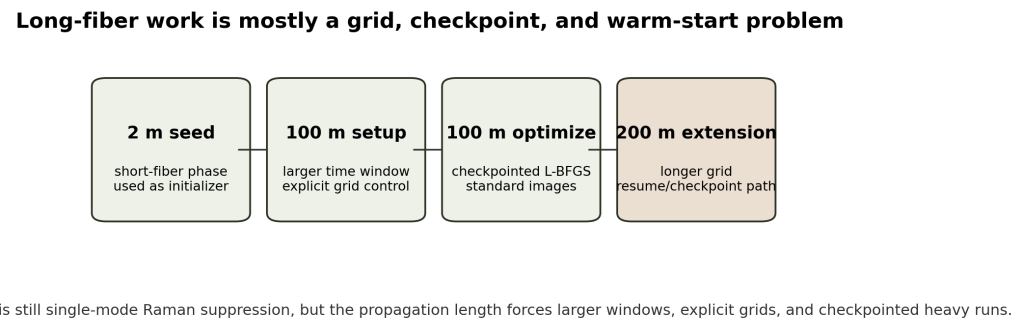


Figure 1: Long-fiber optimization changes the compute and validation workflow more than the physics objective. The solver still optimizes input spectral phase, but the run needs larger windows, explicit grids, warm starts, and checkpoint/restart support.

2 Cost Function and Model

The physical objective is the same Raman-band fraction used in the baseline note:

$$J_{\mathrm{R}}(\phi) = \frac{\sum_{\omega_k \in \mathcal{B}_{\mathrm{Raman}}} |\hat{u}_k(L;\phi)|^2}{\sum_k |\hat{u}_k(L;\phi)|^2}.$$

The input is phase-only:

$$\hat{u}_k(0;\phi) = \hat{u}_{0,k} e^{i\phi_k}.$$

The reported scalar is

$$J_{\mathrm{dB}} = 10 \log_{10} J_{\mathrm{R}}.$$

More negative is better.

The propagation model is the single-mode nonlinear fiber model used elsewhere in the project: dispersion, Kerr nonlinearity, and Raman response in the generalized nonlinear Schrödinger-equation family. This is standard nonlinear fiber-optics modeling territory [1, 2]. The numerical approach follows interaction-picture fiber propagation practice [3].

TL;DR for the Math

Nothing magical happens to the objective when the fiber becomes long. The same fraction is being minimized. What changes is that the pulse has much more distance to disperse, Raman shift, and develop numerical boundary risk. So the main mathematical danger is not redefining the cost; it is using a grid too small to represent the long propagation honestly.

3 Long-Fiber Grid and Checkpoint Strategy

Short-fiber runs can often use compact grids. Long-fiber runs cannot blindly reuse those settings because dispersion walk-off and nonlinear reshaping spread the pulse over a wider temporal interval.

The long-fiber setup therefore honors the requested (N_t, time window) pair instead of letting the short-fiber setup silently resize it.

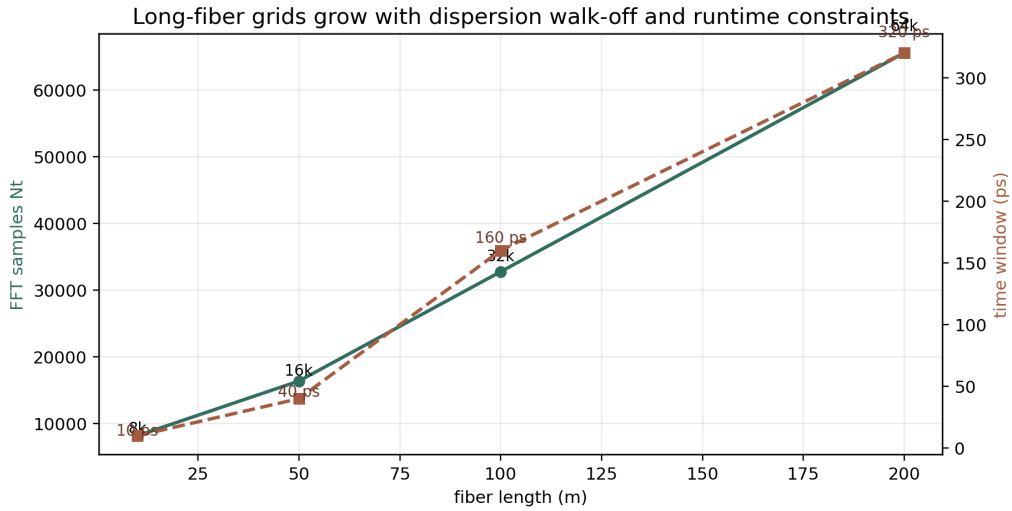


Figure 2: Recommended long-fiber grid ladder used by the research scripts. The 200 m run used $N_t = 65536$ and a 320 ps time window.

The long-fiber optimizer also writes checkpoints. This matters because a 100 m or 200 m solve is expensive enough that losing the process should not lose the research state. The key operational pattern is:

$$\text{warm start} \rightarrow \text{checkpointed L-BFGS} \rightarrow \text{resume if needed} \rightarrow \text{standard images.}$$

Implementation Surface

Component	Role
Problem setup	Builds the explicit long-fiber problem and interpolates warm-start phases across grids.
Optimization driver	Checkpointed L-BFGS driver used for the 100 m and 200 m single-mode runs.
Checkpoint helper	Resume/config-hash guard so long heavy runs can continue safely.
Validation helper	Post-run validation and reporting path for accepted long-fiber artifacts.
Standard-image helper	Regenerates the phase profile, phase diagnostic, optimized heat map, and unshaped-control heat map from saved results.

4 Experimental Method

The long-fiber lane used four checks:

1. propagate the flat/no-shaping control;
2. transfer a strong short-fiber phase profile to the long grid as a warm start;
3. run checkpointed L-BFGS on the long-fiber objective;
4. inspect the standard image set: phase profile, phase diagnostic, optimized heat map, and unshaped-control heat map.

For the 100 m case, the flat control is about $-0.20\,\mathrm{dB}$, the warm-start phase already reaches about $-51.50\,\mathrm{dB}$, and the long optimization reaches the $-55\,\mathrm{dB}$ class. For the 200 m case, the accepted resumed artifact reaches $-55.16\,\mathrm{dB}$.

Provenance and Reproducibility Check

The figures in this note are copied from standard long-fiber image sets: each accepted result has a phase profile, phase diagnostic, optimized evolution heat map, and unshaped-control heat map. The numbers should be read together with the convergence status. A deep final J_{dB} value is not enough by itself; the note also records whether the optimizer converged, what grid was used, and whether the standard images were visually inspected.

TL;DR

The evidence is strong enough to present as a long-fiber simulation milestone. It is not strong enough to call the saved masks globally optimal or lab-ready. The honest claim is: the workflow scales to 100–200 m, produces deep image-backed suppression, and exposes the remaining convergence and actuator readiness gaps.

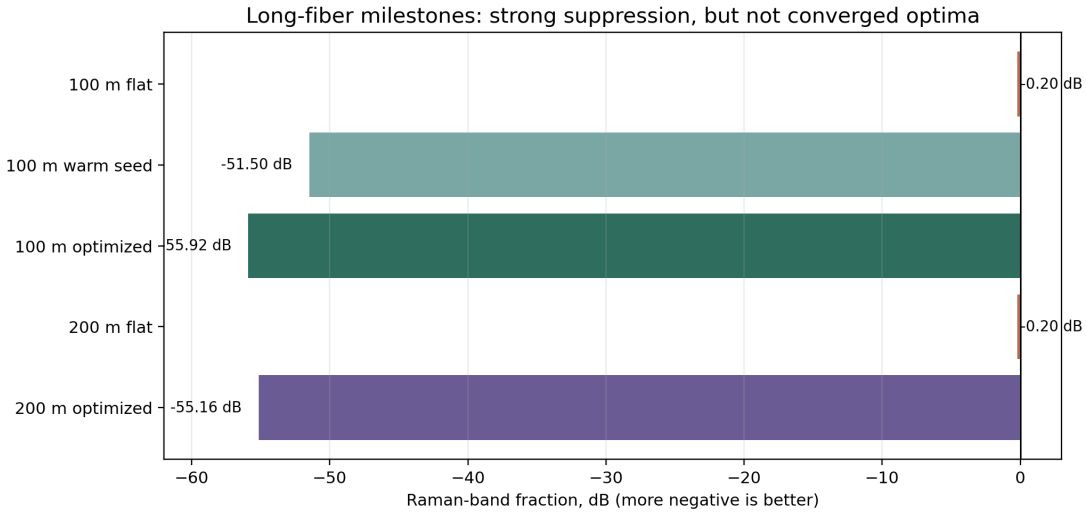
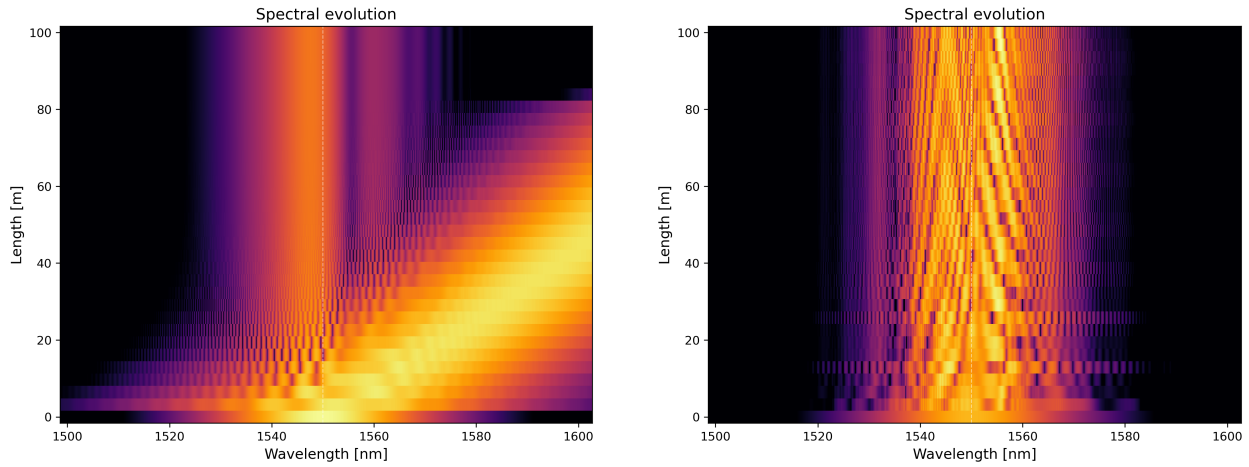


Figure 3: Long-fiber scalar milestones. The 100 m and 200 m optimized results are deep, image-backed suppression results, but the optimizer convergence flags remain false.

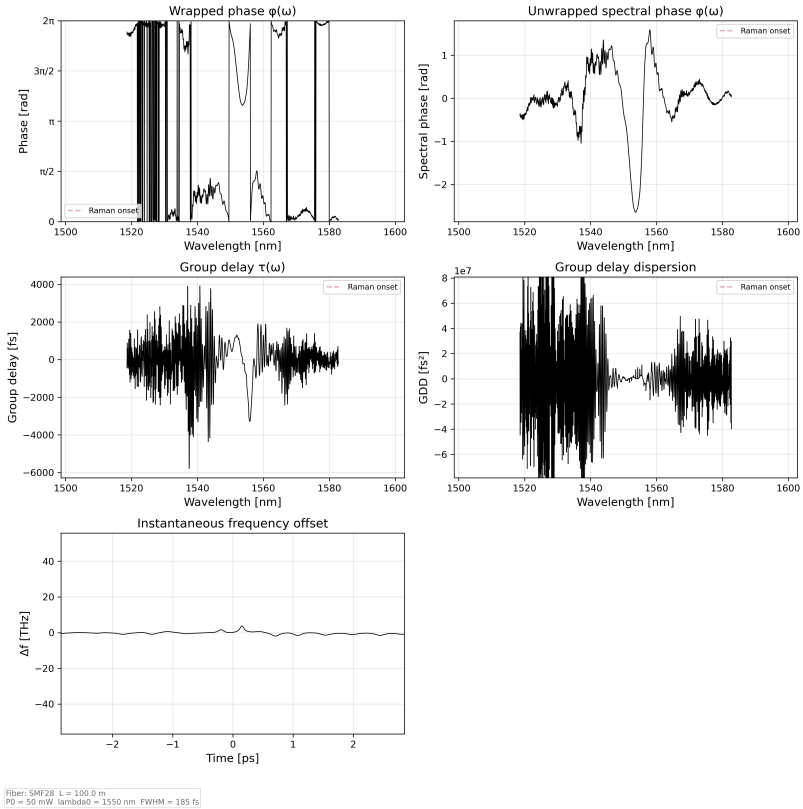
5 100 m Result



100 m no-shaping control.

100 m optimized heat map.

Figure 4: The 100 m comparison is visually strong: the no-shaping control shows heavy Raman-side growth, while the optimized phase keeps the output mostly near the input spectral support.



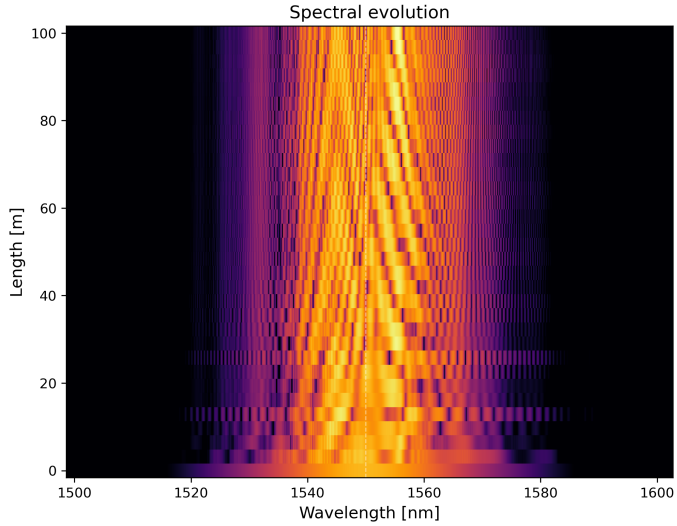
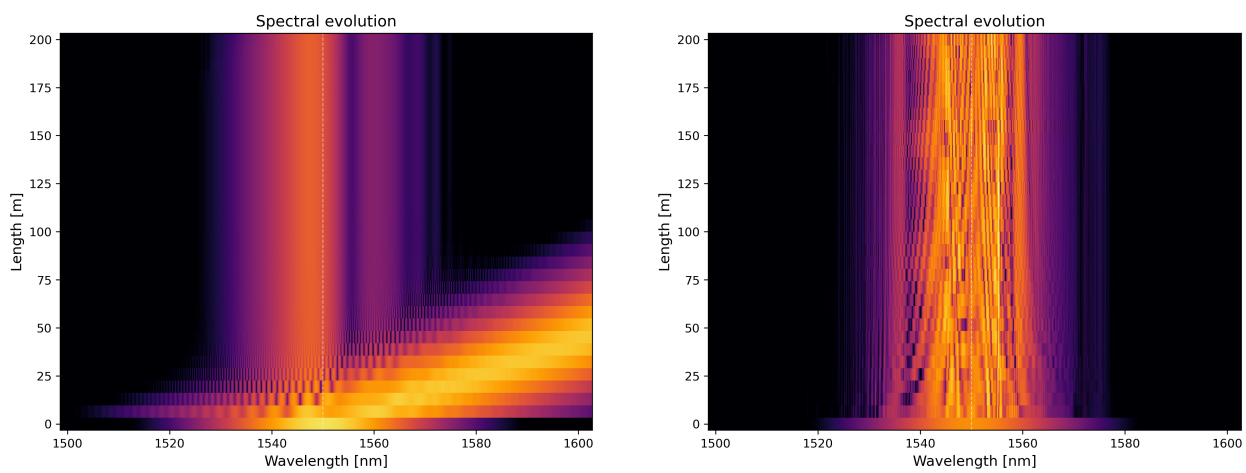


Figure 5: Paired 100 m optimized diagnostic and heat map. The phase is effective but not simple; the diagnostic and group-delay curves should be read as a research result, not as a ready-to-load lab prescription. The matching heat map shows the propagation response produced by that same mask.

6 200 m Result

Quantity	Accepted 200 m resumed result
Fiber length	200.0 m
Power	0.05 W
Grid	$N_t = 65536$, 320 ps
Final Raman-band fraction	−55.1648 dB
Converged flag	false
Residual gradient	0.5649
Standard image set	complete and visually inspected

Table 1: The 200 m result is a completed milestone, not a converged optimum.



200 m no-shaping control.

200 m optimized heat map.

Figure 6: The 200 m standard images show strong suppression of the Raman-side feature relative to the no-shaping control.

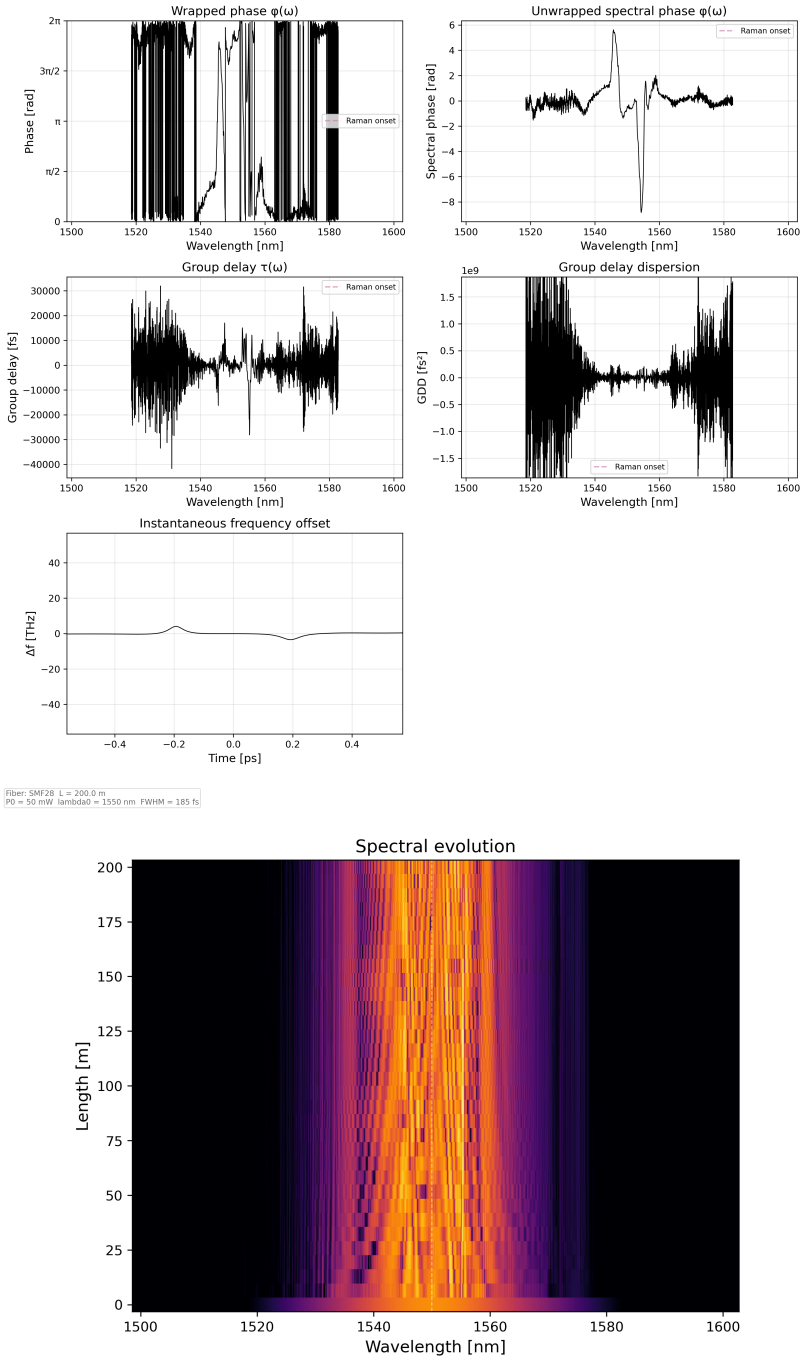


Figure 7: Paired 200 m optimized diagnostic and heat map. The phase is high-structure and not actuator-ready. The main claim is that the long-fiber optimization path reached a deep, visually coherent simulation result.

7 Interpretation

The long-fiber story has three parts. First, phase shaping can still strongly alter Raman transfer far beyond the 2 m baseline. Second, the warm-start seed contains useful structure: before long-fiber

reoptimization, it already lands near the deep-suppression regime. Third, the long-fiber optimized phases are not simple quadratic pre-chirps. The saved quadratic-fit checks found poor weighted polynomial fit quality, which means the useful structure is not just "multiply the dispersion length by a constant."

TL;DR

The long-fiber lane is a proof of reach and workflow: the project can carry phase-only Raman suppression into 100–200 m simulations. It is not yet a proof that the final saved phase is the simplest, most robust, or globally optimal long-fiber waveform.

8 Presentation Capsule

If this note becomes a presentation block, the clean message is:

- The long-fiber result is a scale-up of the same Raman-band objective, not a different cost function.
- The 100 m and 200 m heat maps are strong before/after visuals because the no-shaping controls show obvious Raman growth.
- The compute story matters: explicit grids, warm starts, checkpoints, and standard images are what made the result usable.
- Always include the caveat: these are deep achieved values with non-converged optimizer flags, not final global optima.

9 Limitations

- **Not converged optima.** The accepted 100 m and 200 m results have false convergence flags and nontrivial residual gradients.
- **Single-mode only.** This note does not validate multimode long-fiber optimization.
- **High-structure phases.** The optimized phases are not simple lab-ready actuator prescriptions.
- **Workflow maturity.** The path is usable for research, but not yet a turnkey group-facing long-fiber platform.
- **Grid dependence.** Long-fiber claims must name N_t, time window, length, power, and validation images.

10 Reproduction Capsule

Substantial long-fiber runs belong on the heavy compute machine with Julia threading enabled. The 200 m-style command pattern is:

```
LF100_MODE=fresh \
```

```
LF100_L=200 \
LF100_NT=65536 \
LF100_TIME_WIN=320 \
LF100_RUN_LABEL=longfiber_200m \
LF100_MAX_ITER=15 \
julia -t auto --project=. scripts/research/longfiber/longfiber_optimize_100m.jl
```

The expected durable outputs are:

- checkpoint directory under the long-fiber results tree;
- final result payload;
- phase profile image;
- phase diagnostic image;
- optimized spectral-evolution heat map;
- unshaped-control spectral-evolution heat map.

Before presenting a new long-fiber result, visually inspect the standard image set and record the final dB value, convergence flag, residual gradient, grid, time window, edge fraction, and energy drift.

11 Sources

References

[1] G. P. Agrawal. *Nonlinear Fiber Optics*, 6th ed. Academic Press, 2019. https://www.sciencedirect.com/book/9780128170427/nonlinear-fiber-optics

[2] J. M. Dudley, G. Genty, and S. Coen. "Supercontinuum generation in photonic crystal fiber." *Reviews of Modern Physics*, 78, 1135–1184, 2006. https://doi.org/10.1103/RevModPhys.78.1135

[3] J. Hult. "A Fourth-Order Runge–Kutta in the Interaction Picture Method for Simulating Supercontinuum Generation in Optical Fibers." *Journal of Lightwave Technology*, 25(12), 3770–3775, 2007. https://doi.org/10.1109/JLT.2007.909373